

Course Name- Problem solving using C

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BATCH: 4

Lab 2: Problems based on Data Types, Memory Models & ASCII

Question 1: Employee Salary Analyzer

A company stores employee salary details using different data types.

main.c

```
1  #include <stdio.h>
2  // Input: 101 50000.0 5
3  // Output: Final Salary: 52500.00
4
5  int main(){
6      int e_id;
7      float b_salary, f_salary;
8      char bns_char;
9      int bns_prct;
10
11     scanf("%d %f %c", &e_id, &b_salary, &bns_char);
12
13     bns_prct = bns_char - '0';
14     f_salary = b_salary + (b_salary * bns_prct / 100);
15     printf("Final Salary: %.2f\n", f_salary);
16     return 0;
17 }
```



```
101 50000.0 5
Final Salary: 52500.00
```

```
...Program finished with exit code 0
Press ENTER to exit console.
```

CODE

```
#include <stdio.h>

int main(){
    int e_id;
    float b_salary, f_salary;
    char bns_char;
    int bns_precent;

    scanf("%d %f %c", &e_id, &b_salary, &bns_char);

    bns_precent = bns_char - '0';
    f_salary = b_salary + (b_salary * bns_precent / 100);
    printf("Final Salary: %.2f\n", f_salary);
    return 0;
}
```

Question 2: ASCII-Based Password Validator

```
8
9 #include <stdio.h>
10
11 int main() {
12     char ch;
13     scanf("%c", &ch);
14
15     if(ch >= 65 && ch <= 90) {
16         printf("valid password, size: %lu byte", sizeof(ch));
17     }else{
18         printf("invalid password, size: %lu byte", sizeof(ch));
19     }
20     return 0;
21 }
```

input

A
valid password, size: 1 byte
...Program finished with exit code 0
Press ENTER to exit console.

#include <stdio.h>

```
int main() {
    char ch;
    scanf("%c", &ch);

    if(ch >= 65 && ch <= 90) {
        printf("valid password, size: %lu byte", sizeof(ch));
    }else{
        printf("invalid password, size: %lu byte", sizeof(ch));
    }
    return 0;
}
```